

Shared Borrowers, Shared Stress: The Credit-Line Channel of Contagion

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Barcelona Summer Forum
June 5, 2026

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Motivation

New York Times

NATIONAL EDITION

Rain and snow showers continue from Boston to Philadelphia as a storm moves off shore today. Showers and mountain snow persist in the West. Weather map, Page A24.

company

SATURDAY, MARCH 11, 2023

Prices in Canada may be higher

\$4.00



Labor Market Keeps Riding Its Hot Streak

*U.S. Adds 311,000 Jobs
as Some Sectors Slow*

By LYDIA DePILLIS

The labor market continued its energetic expansion in February, extending a hotter-than-expected streak that has created abundant job opportunities while frustrat-

U.S. TAKES OVER CRUCIAL LENDER IN TECH WORLD

BIGGEST BUST SINCE '08

Investors Batter Stocks of
Other Banks Catering
to Start-Up Clients

The bailout decision

- 13 March 2023: Silicon Valley Bank (SVB) collapses
NYT: “a financial contagion appeared to spread through parts of the financial sector [...] investors dumped stocks of [banks] [...] which cater to start-up clients”
- A failing bank forces a real-time decision: **bail out, or accept failure?**
- To decide, a regulator needs to know — *before markets react*:
 - which banks would be hit, and
 - by how much?

What we do

1. Introduce a **credit-line channel** of bank-to-bank contagion
 - two banks are linked if they lend to the same firms
 - “shared borrowers, shared stress”
2. Turn it into a **spillover matrix** S and bank-level **systemicness**
3. **Compute** S from supervisory microdata (FR Y-14), not market prices
4. Use it as an **ex-ante screen** for the bailout decision

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Why

- market-based measures rely on prices — pick up a crisis *too late*
- supervisory data quantify spillovers *ex ante*

**Mechanism: shared borrowers,
shared stress**

Background: credit lines

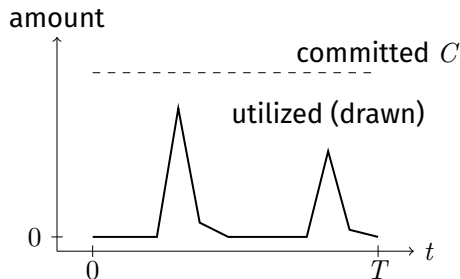
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 - $\sim 73\%$ of public firms
 - worth $\approx 16\%$ of book assets

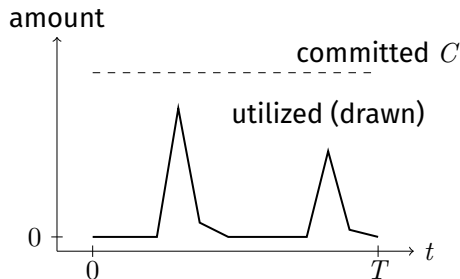
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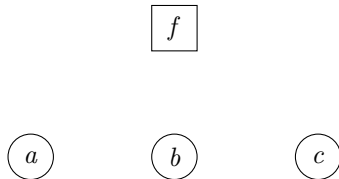


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- **large firms have credit lines with multiple banks** (Greenwald et al., 2025)

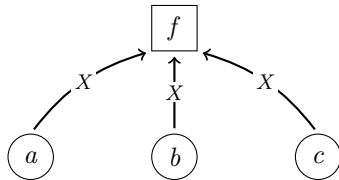


The credit-line channel of contagion



- firm f needs liquidity $3X$, with credit lines at banks a, b, c

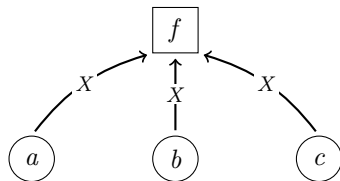
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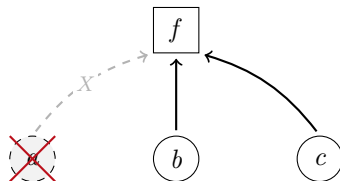
(a) normal times

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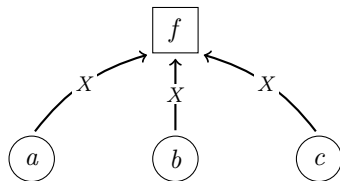
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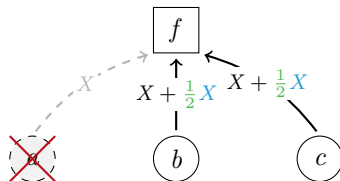
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- firm f needs liquidity $3X$, with credit lines at banks a , b , c
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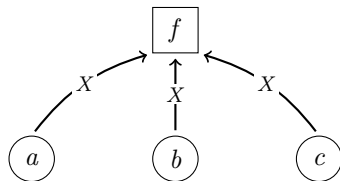
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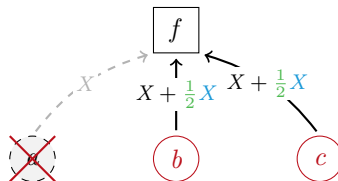
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 - each survivor absorbs its survivor share $\frac{1}{2}$

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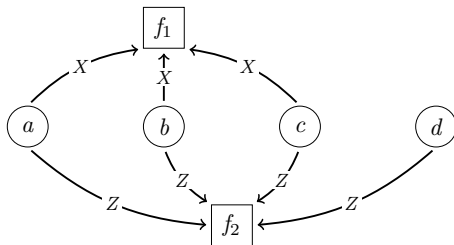
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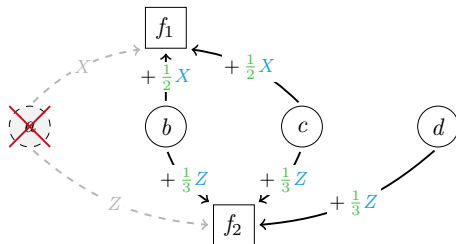
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- firm f needs liquidity $3X$, with credit lines at banks a , b , c
- in normal times, f draws X from each
- if a fails, the **orphaned draw X** must reroute to b and c
 - each survivor absorbs its **survivor share $\frac{1}{2}$**
 - $\implies$ **unexpected liquidity outflow $\frac{1}{2}X$** at b and c

Constructing bank-to-bank spillovers



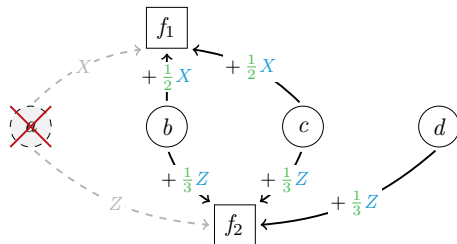
Constructing bank-to-bank spillovers



- the (unnormalized) **spillover** sums the rerouted drawdowns:

$$\tilde{s}_{ab} = \sum_f (\text{survivor share})_{bf}^a \times (\text{orphaned draw})_a^f = \frac{1}{2}X + \frac{1}{3}Z$$

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- normalize, then collect into the **spillover matrix**: $s_{ab} = \tilde{s}_{ab}/B_b$, $S = (s_{ab})$

The channel is real

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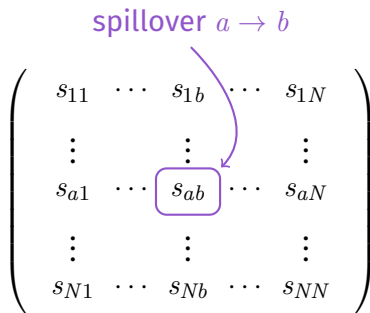
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 - ⇒ the credit-line channel did not fire
 - **Lehman** wasn't subject to FDIC resolution — but since 2008 *all* big banks are

Measuring systemic risk

The object: a spillover matrix S

- **spillover** s_{ab} : how a shock to a hits b — one entry of S

spillover $a \rightarrow b$



The diagram shows a matrix S with rows and columns indexed from 1 to N . The entry s_{ab} is highlighted with a purple box. A purple arrow points from the text "spillover $a \rightarrow b$ " to the entry s_{ab} .

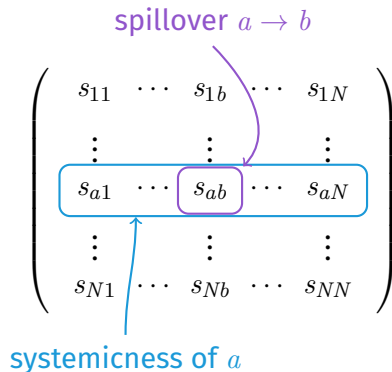
$$\begin{pmatrix} s_{11} & \cdots & s_{1b} & \cdots & s_{1N} \\ \vdots & & \vdots & & \vdots \\ s_{a1} & \cdots & s_{ab} & \cdots & s_{aN} \\ \vdots & & \vdots & & \vdots \\ s_{N1} & \cdots & s_{Nb} & \cdots & s_{NN} \end{pmatrix}$$

Terminology after Greenwood et al. (2015).

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- **systemicness** of a : impact on the whole system — a row of S

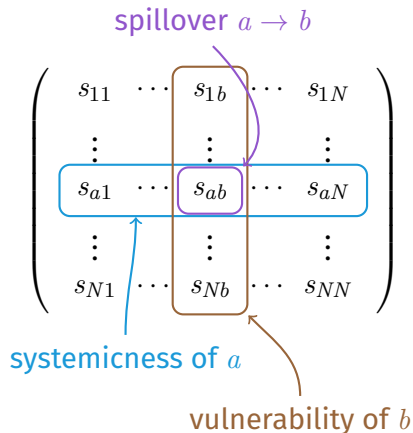
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The object: a spillover matrix S

- **spillover** s_{ab} : how a shock to a hits b — one entry of S
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- **vulnerability** of b : exposure to a system-wide shock — a *column* of S

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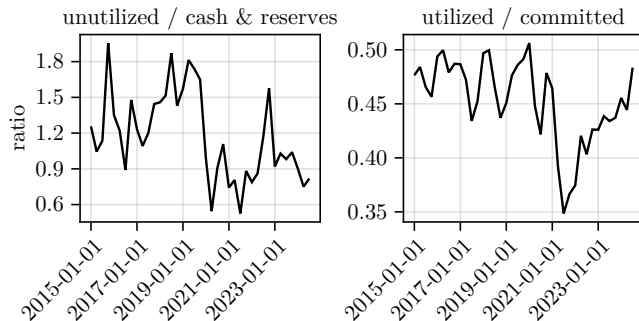
Who can measure it?

	Matrix S ?	Papers	Notes
<i>Reduced-form (equity co-movement)</i>			
equity returns (+ leverage)	aggregates	CoVaR, SRISK, MES	market data needed
equity-return VAR	full matrix	Diebold and Yilmaz (2014)	
<i>Structural (balance-sheet mechanism)</i>			
loans to banks	full matrix	Elsinger et al. (2006)* Denbee et al. (2021)	small
traded securities (fire sales)	aggregates*	Greenwood et al. (2015) Duarte and Eisenbach (2021)	
loans to firms	full matrix	this paper (credit lines)	

Data

- FR Y-14Q — loan-level corporate lending
 - loan amount, type (term loan vs. credit line), borrower characteristics
 - bank holding companies with assets $> \$100\text{bn}$ (≈ 20 banks), 2013–2025
 - build a bank $\times$ sector $\times$ quarter panel of drawn / undrawn credit lines
- FR Y-9C — balance sheets (total assets, cash & reserves, ...)
 - BHCs with assets $> \$3\text{bn}$ ($\gg 20$ banks)

Credit lines and the liquidity buffer



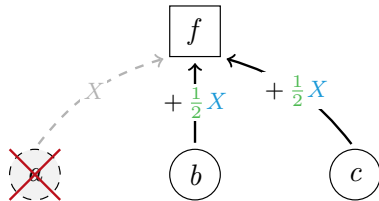
Notes: FR Y-14 and FR Y-9C, 2015Q1–2024Q4; sample restricted to BHCs covered in FR Y-14.

↪ undrawn lines often *outweigh* cash & reserves; utilization typically $< 50\%$

Computing the spillover matrix S

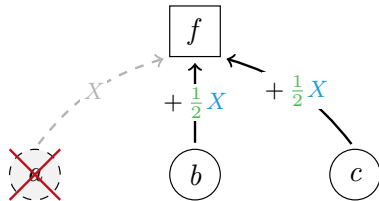
Three ingredients of the spillover s_{ab}

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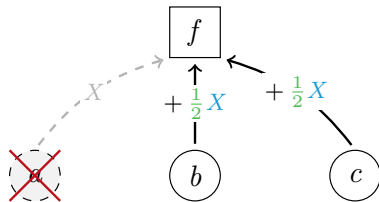
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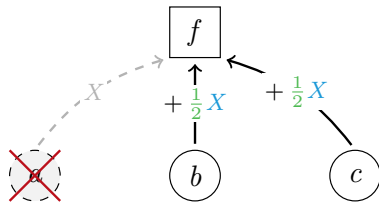
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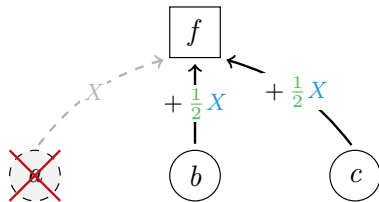
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 - **goal**: infer the **flow** from the **stock**

From stock to flow

Inferring the orphaned draw from the data

1. model a firm's liquidity need as a latent AR(1) process

$$\eta_t = \mu + \rho (\eta_{t-1} - \mu) + \epsilon$$

Inferring the orphaned draw from the data

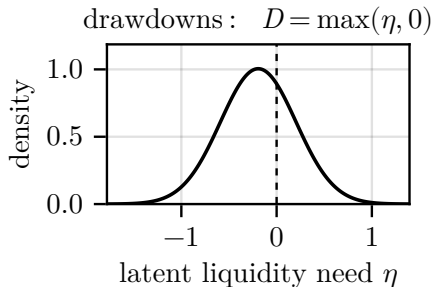
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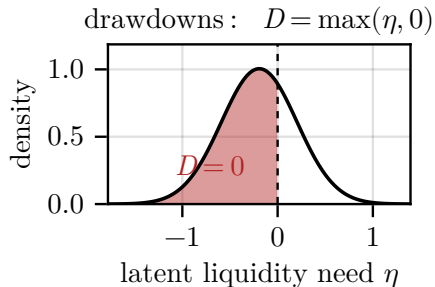
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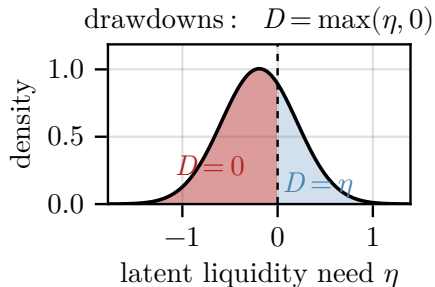


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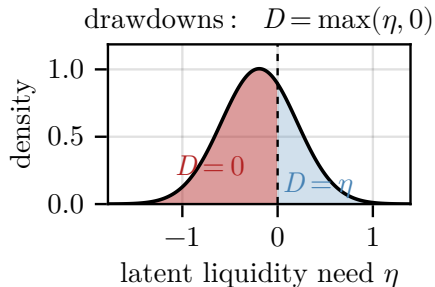
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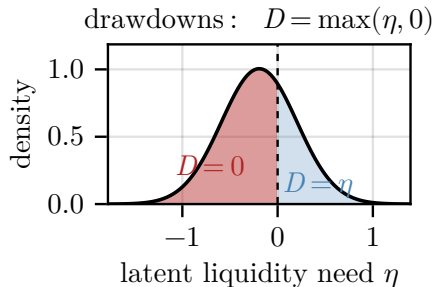
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6. assume $\text{flow} = \phi \cdot \text{stock}$ for each **sector**

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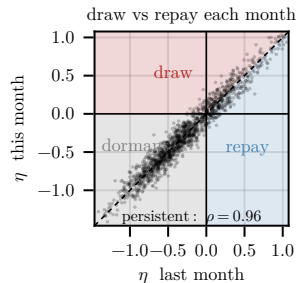
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Inferring the orphaned draw from the data (2)

$$\phi \equiv \mathbb{E}[\text{draw}] / \mathbb{E}[\text{utilized}]$$

- gross draw = $\mathbb{E}[\text{draw}]$

$$\Rightarrow \boxed{\phi \approx 0.165} \text{ / month}$$

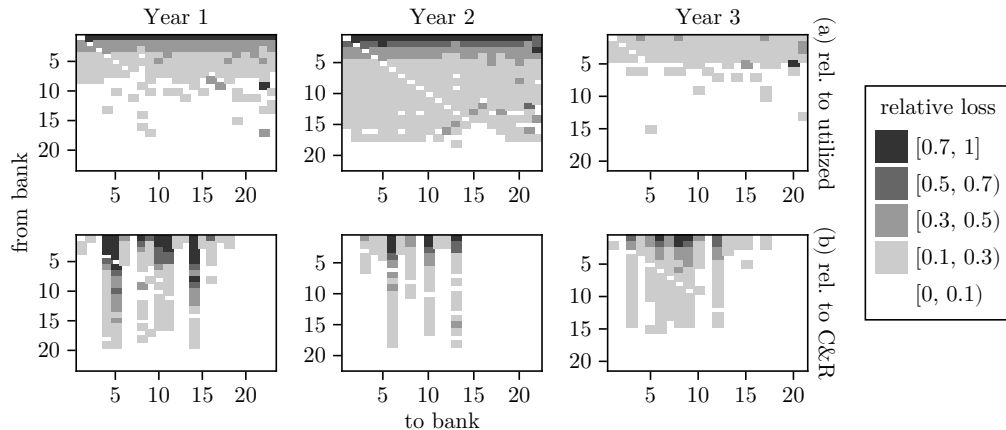


► one ϕ , or many?

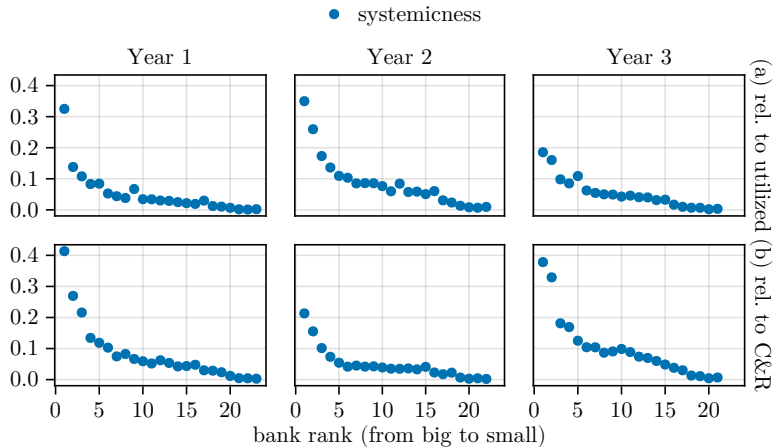
Results

Computed spillover matrix

“additional liquidity outflow at b if a defaulted”



Systemicness: average spillover of a bank's default

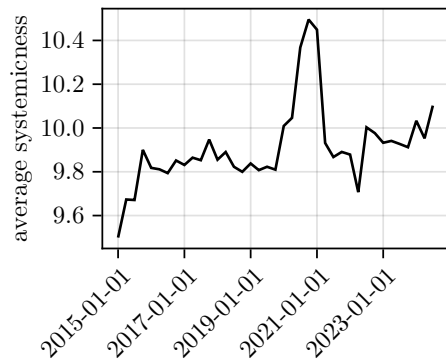


Systemicness is not balance-sheet size

	Systemicness
Committed credit lines	-0.004** (0.001)
Utilized credit lines	0.02741*** (0.00343)
Total assets	0.0895 (0.0515)
<i>N</i>	883
<i>R</i> ²	0.467

Notes: dependent variable is $\log(\text{systemicness})$ at the bank-quarter level; balance-sheet aggregates in billions of USD. Balance-sheet size explains less than half of the variation. ***, **, *: 0.1%, 1%, 5%.

Systemicness over time



Do spillovers price into returns?

An external check — a bank's **exposure** to others' shocks through the network:

$$E_{i,t} = \frac{\sum_{j \neq i} s_{ji,t} \varepsilon_{j,t}^-}{\sum_{j \neq i} s_{ji,t}}, \quad R_{i,t} = \alpha_i + \tau_t + \beta_0 E_{i,t} + e_{i,t}.$$

- $\varepsilon_{j,t}^- \geq 0$: adverse-shock magnitude at bank j (factor-model residual)
- $\hat{\beta}_0 \approx -1.5 \Rightarrow$ markets price the spillover **contemporaneously**

Exposure and contemporaneous returns

	(1)	(2)	(3)	(4)
Exposure	-1.161*** (0.093)	-1.536*** (0.096)	-1.162*** (0.092)	-1.537*** (0.096)
Quarter FE	No	Yes	No	Yes
Bank FE	No	No	Yes	Yes
Observations	2,178	2,178	2,178	2,178
R^2	0.067	0.333	0.061	0.329

Notes: dependent variable is the monthly bank equity return. Exposure is a spillover-weighted average of adverse residualized shocks at other banks. Standard errors in parentheses. ***, **, *: 0.1%, 1%, 5%.

Conclusion

Takeaways

- **Mechanism:** shared credit-line borrowers create a **structural** channel of bank-to-bank contagion
- **Measurement:** supervisory data alone build a spillover matrix and systemicness *ex ante*
- **Policy:** a **screen** for the support-vs-resolution decision — the latent tail that goes live the day no acquirer is found

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- **Measurement:** supervisory data alone build a spillover matrix and systemicness *ex ante*
- **Policy:** a **screen** for the support-vs-resolution decision — the latent tail that goes live the day no acquirer is found
- **Why care, if FDIC takes care of the channel?**
 - **Dormant by accident, not by design** — S prices the day no acquirer is found
 - **Stress $\neq$ default** — a lending cut reroutes through the same S

Backup: Literature on measuring systemic risk

Paper	Scope	Method
Greenwood et al. (2015, JFE)	syst. & vuln.	structural (fire sales)
Duarte and Eisenbach (2021, JF)	syst. & aggr. vuln.	structural (fire sales)
Acharya et al. (2017, RFS)	vulnerability	structural
Brownlees and Engle (2017, RFS)	vulnerability	reduced-form
Adrian and Brunnermeier (2016, AER)	systemicness	reduced-form
Diebold and Yilmaz (2014, JE)	spillover	reduced-form (VAR)
Denbee et al. (2021, JFE)	spillover	structural (interbank)
Elsinger et al. (2006, MS)	spillover	structural (interbank & correlated portfolios)

Terminology follows Greenwood et al. (2015, “vulnerability” vs “systemicness”).

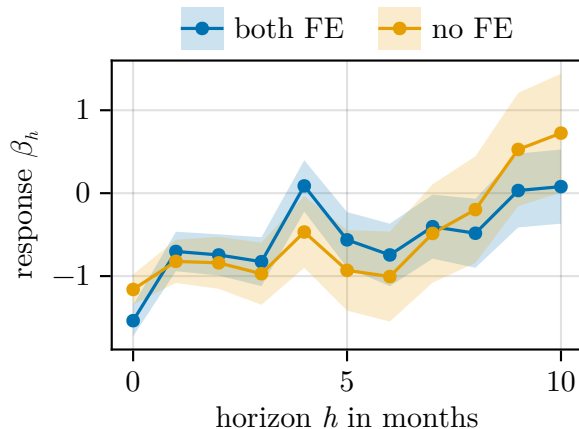
Backup: Credit lines — ubiquity & scope

- **Ubiquity:** $\sim 73\%$ of public firms hold a committed line, $\approx 16\%$ of book assets (Sufi, 2009)
- **Large multi-bank borrowers drive the channel:** small SMEs run near the cap and are gated by lender discretion; large firms hold low utilization in normal times and surge in stress (Chodorow-Reich et al., 2022)

Backup: One ϕ , or many?

- $\phi = \mathbb{E}[\text{draw}]/\mathbb{E}[\text{utilized}]$ is **scale-free** — it depends only on the **zero share** and **persistence**, not on the level of utilization
- so ϕ is **sector-specific in principle** (ϕ_s); we use the large multi-bank-borrower value
- a single ϕ rescales S uniformly $\Rightarrow$ rankings are **regime-invariant** and (we verify) **robust to** ϕ_s — only the levels move

Backup: Dynamic response of returns



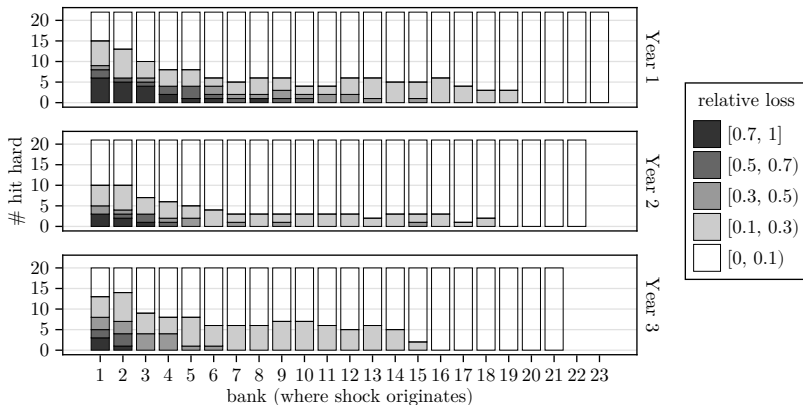
Local-projection estimates: the impact response is negative on impact and attenuates at longer horizons. Shaded bands: 95% CI.

Backup: Summary statistics

Variable	Median	Q25	Q75
<i>FR Y14 (loan-level data)</i>			
committed credit lines	53.4	28.1	106.3
unutilized credit lines	29.6	14.4	72.6
utilized credit lines	24.3	12.1	36.9
<i>FR Y9C (balance sheet data)</i>			
cash and reserves	28.8	10.4	98.4
total assets	272.2	157.1	838.3

Values in billion USD. Sample 2015Q1–2024Q4, restricted to the 23 BHCs observed in at least 30 quarters.

Backup: Systemicness decomposition



Systemicness broken down by component: how individual bilateral spillovers contribute to each bank's aggregate systemicness.

Backup: Factor-model shock construction

At the bank-month level, residualized shocks come from

$$r_{i,t}^e = \alpha + \beta' \mathbf{F}_t + \varepsilon_{i,t},$$

with downside shocks $\varepsilon_{i,t}^- = -\min(\varepsilon_{i,t}, 0) \geq 0$.

- Factors: MKT, SMB, HML, DEF (BAA10Y), TERM (T10Y3M).
- Exposure uses only the negative part of the residual, orthogonal to common equity, credit, and rate factors.

Backup: Confidentiality treatment

- The paper reports the same calendar quarter in three non-consecutive years (Y_1 , Y_2 , Y_3).
- Exact years remain omitted pending Federal Reserve disclosure approval.
- Cross-sectional rankings and magnitudes are the objects for interpretation, not the calendar labels.

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